

Student Course Sales System

Full Design & Architecture Document

8	10	11	7	5
DB Models	API Groups	Pages	Pipeline Stages	Build Phases

Version	1.0
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Status	Draft — Internal Use Only
System	ZyntrixCRM / HRMS Backend (Node.js + MongoDB)
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1. System Overview

The Student Course Sales System is a purpose-built CRM module inside ZyntrixCRM designed for EdTech companies that sell online/offline courses directly to students. The existing generic B2B sales module has been completely replaced with a student-focused pipeline that tracks every student from first contact to course completion.

1.1 What Changed vs. Old System

Old System	New Student System
Generic leads	Student leads with course interest, budget, education level
B2B deal stages	7-stage student enrollment pipeline
No demo tracking	Demo sessions booked, attended, and followed up
No batch/course model	Courses, batches, seats, schedule fully modelled
Manual follow-ups	Automated GAS email triggers at each pipeline stage
No payment tracking	EMI, full-pay, scholarship payment plans with instalment tracking
Fake static data	100% real backend — MongoDB + Express REST APIs

1.2 Technology Stack

Layer	Technology
Backend Runtime	Node.js 18+ / Express 4
Database	MongoDB (Mongoose ODM)
Authentication	JWT (Bearer token) — existing auth middleware
Email Automation	Google Apps Script (GAS) Web App → Gmail API
Frontend	Vanilla HTML + CSS + JS (api.js helper)
Deployment	Render.com (backend) / Static hosting (frontend)
File Uploads	Multer (profile photos, documents)
PDF Generation	ReportLab (Python — reports/exports)

2. 7-Stage Student Pipeline

Every StudentLead moves through exactly 7 stages. Each stage transition can trigger an automated email via GAS and updates the lead's pipelineStage field.

#	Stage	Lead Status	Description	Auto Action
1	New Lead	New Lead	Lead captured from form / import / manual entry	Welcome + course info email
2	Contacted	Contacted	First call / WhatsApp contact made	Follow-up reminder scheduled
3	Demo Scheduled	Demo Scheduled	Demo session booked in system	Demo confirmation email to student
4	Demo Attended	Demo Attended	Student attended demo, interest confirmed	Post-demo nurture email + offer
5	Enrolled	Enrolled	Payment collected, enrollment confirmed	Enrollment confirmation + welcome kit
6	Dropped	Dropped	Student withdrew / not interested	Win-back email after 14 days
7	Completed	Completed	Course completed successfully	Certificate + upsell email

3. Database Models (MongoDB / Mongoose)

All models live under Backend/models/. Each model is registered on the Express server and accessed through authenticated REST APIs.

3.1 StudentLead

Core entity representing one prospective student.

Field	Type / Constraint	Description
_id	ObjectId	Auto-generated primary key
fullName	String req	Student's full name
email	String unique	Primary contact email
phone	String	Mobile number (WhatsApp preferred)
city	String	City / location
educationLevel	String enum	high_school undergraduate graduate working_professional
courseInterest	ObjectId ref:Course	Which course they're enquiring about
budget	Number	Stated monthly/total budget in INR
pipelineStage	String enum	new_lead contacted demo_scheduled demo_attended enrolled dropped completed
source	String enum	website social_media referral cold_call walk_in other
assignedTo	ObjectId ref:User	Sales rep assigned to this lead
lastContactedAt	Date	Timestamp of most recent contact
followUpDate	Date	Next scheduled follow-up date
notes	String	Free-text internal notes
tags	[String]	Custom labels e.g. ['hot', 'scholarship']
createdBy	ObjectId ref:User	Who created the record

3.2 Course

Master catalogue of courses offered by the company.

Field	Type / Constraint	Description
_id	ObjectId	Primary key
title	String req	Course name
slug	String unique	URL-friendly identifier
description	String	Full course description
category	String enum	tech design business marketing other
durationWeeks	Number	Total course length in weeks
price	Number	Standard selling price (INR)
discountPrice	Number	Current discount / offer price
mode	String enum	online offline hybrid
curriculum	[[title, duration]]	Module / topic list
isActive	Boolean default:true	Whether accepting new enrolments
createdBy	ObjectId ref:User	Admin who created course

3.3 Batch

A scheduled run of a course with limited seats.

Field	Type / Constraint	Description
_id	ObjectId	Primary key
course	ObjectId ref:Course req	Which course this batch belongs to
batchCode	String unique	e.g. 'WD-JUN-2026'
startDate	Date	Batch start date
endDate	Date	Batch end date
schedule	String	e.g. 'Mon/Wed/Fri 7–9 PM'
mode	String enum	online offline hybrid
totalSeats	Number	Maximum enrolments allowed
seatsBooked	Number default:0	Auto-incremented on enrolment
instructor	String	Instructor name(s)
meetingLink	String	Zoom / Meet link for online batches
status	String enum	upcoming ongoing completed cancelled

3.4 DemoSession

Tracks each demo / trial class booked for a lead.

Field	Type / Constraint	Description
_id	ObjectId	Primary key
lead	ObjectId ref:StudentLead req	The student who booked
course	ObjectId ref:Course	Course being demoed
scheduledAt	Date req	Demo date & time
mode	String enum	online offline
meetingLink	String	Link if online
venue	String	Venue if offline
conductedBy	String	Instructor / sales rep
attended	Boolean default:false	Did student show up?
feedback	String	Post-demo student feedback
followUpDone	Boolean default:false	Has post-demo follow-up been sent?
createdBy	ObjectId ref:User	Who scheduled the demo

3.5 Enrollment

Confirmed enrolment of a student into a specific batch.

Field	Type / Constraint	Description
_id	ObjectId	Primary key
lead	ObjectId ref:StudentLead req	Enrolled student
batch	ObjectId ref:Batch req	Batch enrolled into
course	ObjectId ref:Course req	Course (denormalized for queries)
enrolledAt	Date default:now	Enrolment timestamp
paymentPlan	String enum	full emi scholarship
totalFee	Number	Agreed total fee
feePaid	Number default:0	Amount paid so far
completionStatus	String enum	active completed dropped
certificateIssued	Boolean default:false	Has certificate been sent?
createdBy	ObjectId ref:User	Sales rep who closed the deal

3.6 Payment

Individual payment transactions (one Enrollment can have many Payments).

Field	Type / Constraint	Description
_id	ObjectId	Primary key
enrollment	ObjectId ref:Enrollment req	Parent enrollment
amount	Number req	Amount paid (INR)
paidAt	Date default:now	Payment timestamp
method	String enum	upi card bank_transfer cash other
transactionId	String	Payment gateway / bank ref
instalmentNumber	Number	1, 2, 3 ... for EMI plans
remarks	String	Internal notes
createdBy	ObjectId ref:User	Who recorded the payment

3.7 FollowUp

Scheduled and completed follow-up log for a lead.

Field	Type / Constraint	Description
_id	ObjectId	Primary key
lead	ObjectId ref:StudentLead req	Lead being followed up
scheduledAt	Date req	When follow-up is due
completedAt	Date	When it was actually done
type	String enum	call whatsapp email meeting
outcome	String enum	no_answer interested not_interested callback enrolled
notes	String	What was discussed
nextFollowUp	Date	Date of next follow-up if needed
createdBy	ObjectId ref:User	Sales rep who created the follow-up

3.8 SalesTarget

Monthly targets per sales rep — used for performance dashboard.

Field	Type / Constraint	Description
_id	ObjectId	Primary key
user	ObjectId ref:User req	Sales rep
month	Number	1–12
year	Number	e.g. 2026
targetLeads	Number	Lead generation target
targetEnrollments	Number	Enrollment closure target
targetRevenue	Number	Revenue target (INR)
achievedLeads	Number default:0	Actual leads so far
achievedEnrollments	Number default:0	Actual enrollments so far
achievedRevenue	Number default:0	Actual revenue so far

4. REST API Route Groups

All routes are protected by JWT auth middleware. Base URL: `/api/sales/...` The Express router file is `Backend/routes/salesRoutes.js`.

4.1 Student Leads

Method	Endpoint	Description
GET	<code>/leads</code>	List all leads (filter by stage, assignedTo, source, date range)
POST	<code>/leads</code>	Create a new student lead
GET	<code>/leads/:id</code>	Get single lead with populated fields
PATCH	<code>/leads/:id</code>	Update lead fields (incl. pipelineStage)
DELETE	<code>/leads/:id</code>	Soft-delete / archive lead
PATCH	<code>/leads/:id/stage</code>	Move lead to new pipeline stage (triggers email)
GET	<code>/leads/stats</code>	Aggregate counts by stage, source, rep

4.2 Courses

Method	Endpoint	Description
GET	<code>/courses</code>	List all active courses
POST	<code>/courses</code>	Create new course (admin only)
GET	<code>/courses/:id</code>	Get course with curriculum
PATCH	<code>/courses/:id</code>	Update course details
DELETE	<code>/courses/:id</code>	Deactivate course (soft delete)

4.3 Batches

Method	Endpoint	Description
GET	<code>/batches</code>	List batches (filter by course, status, date)
POST	<code>/batches</code>	Create new batch
GET	<code>/batches/:id</code>	Get batch with enrolled count
PATCH	<code>/batches/:id</code>	Update batch details
DELETE	<code>/batches/:id</code>	Cancel batch
GET	<code>/batches/:id/enrollments</code>	List all enrollments for a batch

4.4 Demo Sessions

Method	Endpoint	Description
GET	/demos	List all demos (filter by date, attended, lead)
POST	/demos	Schedule a new demo session
GET	/demos/:id	Get demo details
PATCH	/demos/:id	Update demo (mark attended, add feedback)
DELETE	/demos/:id	Cancel/delete demo
POST	/demos/:id/send-reminder	Trigger GAS reminder email to student

4.5 Enrollments

Method	Endpoint	Description
GET	/enrollments	List all enrollments (filter by batch, status)
POST	/enrollments	Create enrollment (auto-increments batch seats)
GET	/enrollments/:id	Get enrollment with payments
PATCH	/enrollments/:id	Update enrollment status / completion
DELETE	/enrollments/:id	Cancel enrollment (decrements seats)

4.6 Payments

Method	Endpoint	Description
GET	/payments	List all payments (filter by enrollment, date)
POST	/payments	Record a new payment transaction
GET	/payments/:id	Get payment details
PATCH	/payments/:id	Correct payment record
DELETE	/payments/:id	Void/delete payment
GET	/payments/summary	Total collected, pending, overdue by enrollment

4.7 Follow-Ups

Method	Endpoint	Description
GET	/followups	List follow-ups (filter by due today, overdue, lead)
POST	/followups	Schedule a new follow-up
GET	/followups/:id	Get follow-up details
PATCH	/followups/:id	Mark completed, add outcome & next follow-up
DELETE	/followups/:id	Delete follow-up
GET	/followups/today	All follow-ups due today for current user

4.8 Sales Targets

Method	Endpoint	Description
GET	/targets	List all targets (current user or all — admin)
POST	/targets	Set target for a rep for a month
PATCH	/targets/:id	Update target values
GET	/targets/dashboard	Current month target vs achieved for all reps

4.9 Reports & Analytics

Method	Endpoint	Description
GET	/reports/pipeline	Lead count by stage (funnel chart data)
GET	/reports/revenue	Revenue collected by month / course / rep
GET	/reports/conversion	Lead-to-enrollment conversion rates
GET	/reports/demos	Demo scheduled vs attended rates
GET	/reports/rep-performance	Per-rep stats: leads, demos, enrollments, revenue

4.10 Bulk / Import

Method	Endpoint	Description
POST	/leads/import	Bulk import student leads from Excel (.xlsx)
GET	/leads/export	Export all leads to Excel
POST	/courses/import	Bulk import course catalogue

5. Frontend Pages

All pages live under Frontend/modules/Sales/. Each page uses the shared api.js helper and follows the existing ZyntrixCRM design system (dark navy + indigo).

#	File	Status	Page Name	Key Features
5.1	leads.html	NEW	Student Leads Pipeline	- Kanban view — 7 columns, one per pipeline stage - Drag-and-drop card to move lead between stages - Quick-add lead form in slide-over panel - Filter bar: stage, source, assigned rep, date range - Click card → opens detail drawer
5.2	lead_detail.html	NEW	Lead Detail & Timeline	- Full lead profile: personal info, course interest, budget - Activity timeline: calls, demos, payments, stage changes - Inline follow-up scheduler with outcome tracker - One-click 'Book Demo' button → creates DemoSession - One-click 'Enroll' button → opens enrollment modal
5.3	courses.html	NEW	Course Catalogue	- Card grid of all active courses with price, duration, mode badges - Create/edit course slide-over form with curriculum builder - Batch count and available seats shown per course card - Search and category filter
5.4	batches.html	NEW	Batch Manager	- Table view: batch code, dates, seats filled/total, status chip - Create batch form linked to course dropdown - View batch → shows enrolled students list - Bulk-mark batch status (ongoing → completed)
5.5	demos.html	NEW	Demo Sessions Calendar	- Month/week calendar view of scheduled demos - List view with attended / not-attended toggle - Send reminder email button per demo - Quick 'Mark Attended' + feedback notes popup - KPI bar: total scheduled, attended rate, post-demo conversion
5.6	enrollments.html	NEW	Enrollments	- Table: student name, course, batch, plan, fee paid/total - Color-coded payment status: Paid / Partial / Overdue - Open enrollment → view payment history + record new payment - Issue certificate button (updates flag, triggers email) - Export to Excel button
5.7	payments.html	NEW	Payment Register	- Full transaction log with method, amount, date, instalment number - Search by enrollment, date range, method - Running total cards: collected today / this month / overdue - Void payment button (admin only)
5.8	followups.html	NEW	Follow-Up Tracker	'Due Today' list at top — highlighted in orange if overdue - Mark complete with outcome dropdown and next follow-up picker - Auto-advance pipeline stage on 'enrolled' outcome selection - Per-rep filter for team lead view
5.9	targets.html	REWORK	Sales Targets Dashboard	- Admin sets monthly targets per rep - Progress bars: leads / demos / enrollments / revenue vs target - Team leaderboard sorted by revenue achievement % - Month picker to view historical target performance

#	File	Status	Page Name	Key Features
5.10	reports.html	NEW	Reports & Analytics	• Pipeline funnel chart (lead counts by stage) • Revenue bar chart by month with course breakdown • Conversion rate table: source → lead → demo → enrollment • Rep performance comparison table • Date range picker + export to PDF button
5.11	import.html	REWORK	Import / Export	• Drag-and-drop Excel upload for bulk lead import • Column mapping UI (existing import framework re-used) • Download sample template button • Export leads / enrollments to Excel

6. GAS Email Automation Triggers

Google Apps Script (GAS) handles all outbound emails to students. The Node.js backend calls the GAS Web App with a JSON payload via HTTP POST. GAS uses MailApp.sendEmail() (100 emails/day limit on free Gmail accounts).

Trigger Name	When Fired	Payload Fields	Email Content
sendWelcome	Lead created (pipelineStage: new_lead)	{ name, email, courseTitle, contactRep, contactPhone }	Welcome email with course brochure link and next steps
sendDemoConfirmation	Demo session scheduled	{ name, email, courseTitle, demoDate, demoTime, meetingLink, venue }	Demo booking confirmation with join/location details
sendDemoReminder	24 hours before demo (or manual trigger)	{ name, email, courseTitle, demoDate, demoTime, meetingLink }	Day-before reminder with a countdown and link
sendEnrollmentConfirmation	Enrollment record created	{ name, email, courseTitle, batchCode, startDate, totalFee, paymentPlan }	Enrollment welcome + batch schedule + payment plan details
sendCertificate	Enrollment marked completed + certificateIssued=true	{ name, email, courseTitle, completionDate, certificateUrl }	Congratulations email with certificate download link and upsell offer

6.1 GAS Handler Structure (Node.js side)

In Backend/utils/studentEmails.js — each trigger function calls callGasEmail() with the action name and payload. The GAS script dispatches based on action field.

```
// candidateEmails pattern re-used for student leads
async function notifyDemoConfirmation(demo, lead, course) {
  return callGasEmail('sendDemoConfirmation', {
    name: lead.fullName,
    email: lead.email,
    courseTitle: course.title,
    demoDate: demo.scheduledAt.toDateString(),
    demoTime: demo.scheduledAt.toTimeString().slice(0,5),
    meetingLink: demo.meetingLink || '',
    venue: demo.venue || ''
  });
}
```

7. Build Phases

The system is built in 5 sequential phases. Each phase has a specific deliverable that can be tested independently before moving to the next phase.

Phase

1

Backend Foundation

Est. 1–2 days

- ✓ Create all 8 Mongoose models with validation
- ✓ Set up salesRoutes.js with all route groups
- ✓ Wire routes in server.js under /api/sales
- ✓ Write salesController.js — CRUD for all models
- ✓ Test all endpoints with Postman / Thunder Client

Phase

2

Pipeline & Lead Management

Est. 1–2 days

- ✓ Build leads.html — Kanban pipeline board
- ✓ Build lead_detail.html — timeline + actions
- ✓ Implement pipelineStage change logic with history log
- ✓ Add bulk import from Excel (re-use existing import framework)
- ✓ Add lead export to Excel

Phase

3

Courses, Batches & Demos

Est. 1–2 days

- ✓ Build courses.html — catalogue with CRUD
- ✓ Build batches.html — batch manager with seat tracking
- ✓ Build demos.html — calendar view + mark-attended flow
- ✓ Auto-update lead pipelineStage when demo is booked / attended
- ✓ Seat count auto-increment/decrement on enrollment create/cancel

**Phase
4****Enrollments, Payments & GAS Emails**

Est. 2 days

- ✓ Build enrollments.html — enrollment table + payment history
- ✓ Build payments.html — transaction register
- ✓ Implement EMI instalment tracking and overdue flags
- ✓ Create Backend/utills/studentEmails.js with all 5 GAS triggers
- ✓ Update GAS script with new handlers (sendWelcome, sendDemoConfirmation, etc.)
- ✓ Issue certificate flow — flag + certificate email

**Phase
5****Follow-Ups, Targets & Reports**

Est. 2 days

- ✓ Build followups.html — due-today list + outcome tracker
- ✓ Build targets.html — rep target vs achieved dashboard
- ✓ Build reports.html — funnel, revenue, conversion charts
- ✓ Implement /api/sales/reports/* aggregation endpoints
- ✓ Performance optimisation: add indexes to frequently queried fields
- ✓ End-to-end QA pass across all pages and API routes

8. Summary Statistics

Component	Count	Detail
MongoDB Models	8	StudentLead, Course, Batch, DemoSession, Enrollment, Payment, FollowUp, SalesTarget
API Route Groups	10	Leads, Courses, Batches, Demos, Enrollments, Payments, Follow-Ups, Targets, Reports, Import/Export
Total API Endpoints	38+	Full CRUD + stats + export endpoints across all groups
Frontend Pages	11	9 new pages + 2 reworked pages (targets, import)
GAS Email Triggers	5	Welcome, Demo Confirmation, Demo Reminder, Enrollment Confirmation, Certificate
Pipeline Stages	7	New Lead → Contacted → Demo Scheduled → Demo Attended → Enrolled → Dropped → Completed
Build Phases	5	Foundation → Lead Management → Courses & Demos → Payments → Reports (est. 8–10 days total)

This document is the authoritative design reference for the ZyntrixCRM Student Course Sales System. All development should follow the models, API contracts, and page specifications defined here. Update this document when the design changes.

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